



Raspberry Pi Compute Module 5

A Raspberry Pi for deeply embedded applications

Colophon

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Release	3
Build date	08/06/2026

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Table of Contents

1. Introduction	4
1.1. Connectors	4
1.2. Compatibility	4
1.3. Features	5
2. Interfaces	6
2.1. Wireless	6
2.1.1. Wi-Fi disable (WL_nDisable)	6
2.1.2. Bluetooth disable (BT_nDisable)	6
2.2. Ethernet	7
2.2.1. Connector and design guidance	7
2.2.2. Status LEDs and sync output	7
2.3. PCIe Gen 2	8
2.3.1. Routing guidance	8
2.3.2. Required signals	8
2.4. USB interfaces	8
2.4.1. USB 3.0 (SuperSpeed)	8
2.4.2. USB 2.0 (High-Speed)	8
2.5. Video and display interfaces	9
2.5.1. Dual HDMI 2.0	9
2.5.2. MIPI (CSI and DSI)	9
2.6. I2C interfaces	10
2.6.1. MIPI I2C bus (SDA0 and SCL0)	10
2.6.2. HAT EEPROM identification I2C bus (ID_SD and ID_SC)	10
2.7. SDIO (CM5Lite only)	10
2.8. Debug UART	11
2.9. GPIO	11
2.9.1. Alternative function assignments	11
2.9.2. Alternative GPIO functions	12
2.9.3. Camera GPIOs (CAM_GPIO)	13
2.10. Status LEDs (LED_nACT and LED_nPWR)	13
2.11. Fan control (Fan_PWM and Fan_Tacho)	13
2.12. Power management and control	13
2.12.1. USB-C signals (CC0 and CC1)	13
2.12.2. System control signals (PMIC_EN, PWR_BUT, VBAT, nRPI_BOOT, and EEPROM_nWP)	14
3. Power	15
3.1. Power-up sequencing	15
3.2. Power-down sequencing	15
3.3. Power consumption	15
3.4. Regulator outputs	15
4. Specifications	16
4.1. Mechanical specifications	16
4.1.1. PCB dimensions	16
4.1.2. Wireless antenna	17
4.2. Pinout	17
4.2.1. Pin guidelines	22
4.2.2. Differential pairs	23
4.3. Electrical specifications	25
4.3.1. Absolute maximum ratings	25
4.3.2. DC characteristics	25
4.3.3. Current consumption	26
4.4. Thermal characteristics	27
4.5. Mean time between failure (MTBF)	27
5. Troubleshooting	28
5.1. Hardware power rails	28

5.2. Bootloader firmware	28
5.3. EEPROM management and firmware updates	28
6. Ordering information	29
6.1. Order quantity and packaging	29
6.2. Part number codes	29
6.3. Product variants	29
Appendix A. Test Points	31
A.1. Test point map	31
A.2. Test point connections for power and programming	33
Appendix B. CM4 and CM5 differences	34
B.1. Pinout changes	34
B.1.1. Per-pin differences	34
B.1.2. Summary of functional and hardware changes	35
B.2. Track lengths	35
B.3. Power budget	35
Appendix C. Documentation history	36

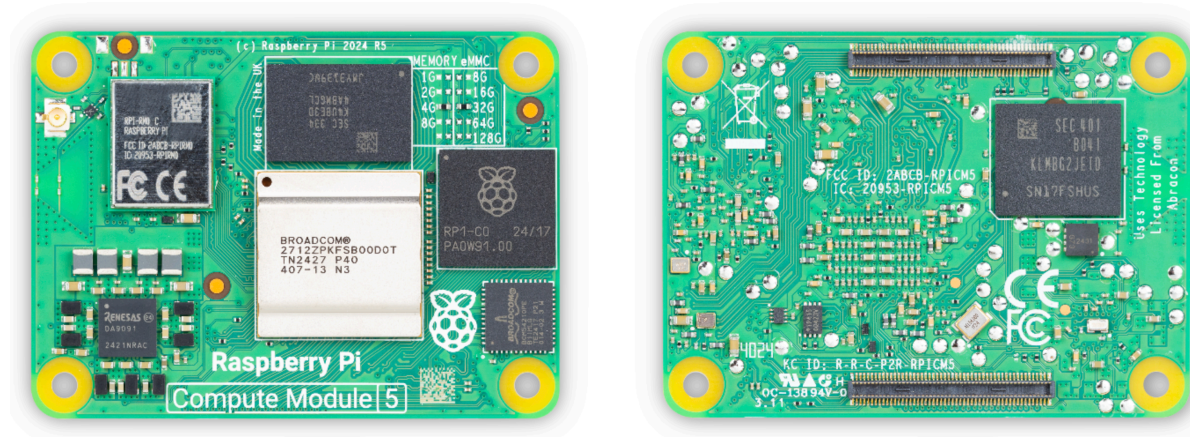
1. Introduction

Raspberry Pi Compute Module 5 (CM5) is a System on Module (SoM) designed to deliver the functionality of Raspberry Pi in a compact and flexible form factor suited to embedded and industrial applications. It enables developers and system designers to leverage the Raspberry Pi hardware and software stack in their own custom systems and designs.

CM5 includes a processor, memory (RAM), eMMC flash storage, and supporting power circuitry. It also provides I/O interfaces beyond those available on standard Raspberry Pi boards, offering expanded options for more complex systems and designs.

Figure 1.

The front (left) and back (right) of Raspberry Pi Compute Module 5 (CM5)



For support documentation for CM5, see the [Compute Module](https://www.raspberrypi.com) section of <https://www.raspberrypi.com>. You can also post a question to the [Forum](#).

1.1. Connectors

CM5 includes two 100-pin high-density connectors, providing access to nearly all CM5 interfaces. Together, these connectors transmit power, data, and control signals to a carrier board. The top connector on CM5 contains pins 1 to 100; the bottom connector of CM5 contains pins 101 to 200. For information about each pin's assignment, see [Section 4.2. Pinout](#).

CM5 has a companion carrier board, the Raspberry Pi Compute Module 5 IO Board (CM5IO) board, which is designed to expose and enable CM5 interfaces. You can also design your own carrier board based on the CM5IO board. The CM5IO design files are freely available. For detailed specifications and pinout information about CM5IO, see the [Compute Module 5 IO board documentation](#).

1.2. Compatibility

CM5 and CM5Lite are mostly compatible with the previous generation of Compute Module. This means that CM5 can be used in many existing designs and carrier boards with minimal changes. For a list of specific differences between CM5 and CM4, see [Appendix B. CM4 and CM5 differences](#).

Note

The previous generation of Compute Module (CM4) is still for sale and will remain in production until at least January 2034.

CM5 connects to carrier boards through its two 100-pin connectors. The main change in the pin layout (pinout) compared to the previous Compute Module is the addition of support for two USB 3.0 ports. For a list of pin differences between CM5 and CM4, see [Appendix B.1. Pinout changes](#).

1.3. Features

The design of CM5 is loosely based on Raspberry Pi 5. For cost-sensitive applications, CM5 is also available without the eMMC storage; this variant is called Raspberry Pi Compute Module 5 Lite (CM5Lite). Unless otherwise stated, within this document, CM5 also refers to CM5Lite.

Key features of CM5 are as follows:

- **High-performance SoC.** Broadcom BCM2712 quad-core Cortex-A76 (ARMv8) 64-bit processor running at 2.4 GHz.
- **Compact module design.** Small footprint of 55 mm × 40 mm × 4.7 mm module with four M2.5 mounting holes.
- **Video decoding.** Hardware-accelerated 4Kp60 HEVC video decoder.
- **Graphics support.** OpenGL ES 3.1 and Vulkan 1.2 for modern GPU acceleration.
- **Memory options.** Available with 2 GB, 4 GB, 8 GB, or 16 GB LPDDR4x-4267 SDRAM with ECC support. For more information about memory options, see [Section 6. Ordering information](#).
- **Flash storage.** Fast onboard eMMC flash storage with the following options:
 - **Speed.** An eMMC bandwidth of up to 400 MB/s, which is four times faster than previous compute modules.
 - **Storage.** Options for 16 GB, 32 GB, or 64 GB eMMC flash memory (for **CM5**), or no eMMC flash memory (**CM5Lite**). For more information about storage options, see [Section 6. Ordering information](#).
- **Additional SDIO interface for CM5Lite.** One SDIO 2.0 interface to provide external storage or peripheral expansion in place of onboard eMMC (CM5Lite only).
- **Optional certified wireless module.** Option (see [Section 6. Ordering information](#)) for certified radio module with:
 - Dual-band Wi-Fi (2.4 GHz and 5.0 GHz IEEE 802.11 b/g/n/ac).
 - Bluetooth 5.0 with BLE.
 - On-board electronic antenna switch that allows selection between PCB trace or external antenna.
- **Wired networking.** Integrated Gigabit Ethernet PHY with IEEE 1588 precision time protocol support.
- **PCIe expansion.** One-lane PCIe Gen 2 (5 Gb/s) host interface for high-speed peripherals.
- **USB connectivity.** USB options for both High-Speed and SuperSpeed peripherals:
 - One USB 2.0 high-speed port.
 - Two USB 3.0 (SuperSpeed) ports, supporting simultaneous 5 Gb/s data transfer.
- **Flexible GPIO and peripheral support.** Up to 30 GPIOs, supporting 1.8 V or 3.3 V signalling, with multiple peripheral interfaces:
 - Up to five UART
 - Up to five I2C
 - Up to five SPI
 - One SDIO interface
 - One DPI (parallel RGB display)
 - One I2S
 - Up to four PWM channels
 - Up to three GPCLK outputs
- **Dual HDMI outputs.** Two HDMI 2.0 ports, each supporting up to 4Kp60 output simultaneously.
- **Dual 4-lane MIPI interfaces.** Two MIPI ports supporting both DSI (display port) and CSI-2 (camera port) functionality.
- **Power input.** Single 5 V power input with USB power delivery support for up to 5 A at 5 V.
- **Real-time clock (RTC).** Integrated RTC powered by an external battery for timekeeping when offline.
- **Fan control.** Dedicated 2-pin fan control with PWM for active thermal management.

2. Interfaces

CM5 includes a range of interfaces (physical connectors, control signals, and configuration mechanisms) to support diverse applications, from high-speed storage and networking to wireless communication, display outputs, and flexible GPIO expansion. These interfaces allow you to build connected and adaptable embedded systems. The following sections provide technical information on each available interface, including configuration options, routing guidelines, and design considerations.

2.1. Wireless

CM5 supports both Wi-Fi and Bluetooth functionality, allowing developers and system designers to flexibly manage wireless connectivity for a range of applications.

The wireless interfaces on CM5 are provided by the Cypress CYW43455 silicon, supporting both:

- 2.4 GHz and 5.0 GHz IEEE 802.11 b/g/n/ac Wi-Fi.
- Bluetooth 5.0 and BLE.

You can enable and disable these wireless functions independently as required. For example, in kiosk deployments, a service engineer might temporarily enable wireless to perform updates, then disable it for security and regulatory compliance.

CM5 has an on-board PCB antenna that should be positioned away from conductive materials, such as metal or ground planes. For more information, see [Section 4. Specifications](#). Alternatively, you can connect an external antenna through a standard U.FL connector. For the location of the connector, see the [CM5 mechanical diagram](#) in [Section 4.1.1. PCB dimensions](#).

Antenna selection (internal or external) is configured at boot time using the `config.txt` file. This selection can't be changed during operation. To select the antenna, append one of the following lines to `config.txt`:

- `dtparam=ant1` selects the internal PCB antenna.
- `dtparam=ant2` selects the external antenna through a U.FL connector.

Raspberry Pi Ltd offers a certified antenna kit for use with CM5. If you use a third-party antenna, you must obtain your own separate certification because Raspberry Pi Ltd doesn't support certification with non-approved antennas.

Important

Raspberry Pi Ltd doesn't assist with certification for third-party antennas.

To support power savings and regulatory usage requirements, two control pins, wireless (Wi-Fi) disable (`WL_nDisable`) and Bluetooth disable (`BT_nDisable`), allow hardware-level shut down of Wi-Fi and Bluetooth, respectively. These pins are reserved on Compute Modules without wireless functionality.

2.1.1. Wi-Fi disable (`WL_nDisable`)

The `WL_nDisable` pin indicates the enable/disable state of Wi-Fi and may also be used to disable Wi-Fi. This pin may only be driven low; it can't be driven high. The software driver drives it high internally when required.

- If the pin is high (logic 1), Wi-Fi is powered up. If Wi-Fi is enabled after being disabled, you must reinitialise the Wi-Fi driver.
- When driven or tied low (logic 0), the pin prevents Wi-Fi from powering up, helping to reduce power consumption or meet requirements to physically disable Wi-Fi.

2.1.2. Bluetooth disable (`BT_nDisable`)

The `BT_nDisable` pin indicates the enable/disable state of Bluetooth and may also be used to disable Bluetooth. This pin may only be driven low; it can't be driven high. The software driver drives it high internally when required.

- If the pin is high (logic 1), Bluetooth is powered up. If Bluetooth is enabled after being disabled, you must reinitialise the Bluetooth driver.
- When driven or tied low (logic 0), the pin prevents Bluetooth from powering up, helping to reduce power consumption or meet requirements to physically disable Bluetooth.

2.3. PCIe Gen 2

CM5 has an internal PCIe 2.0 host controller, offering high-speed expansion options for NVMe storage, networking cards, and other peripherals. Operation in PCIe Gen 3.0 mode is possible in some cases, but is unsupported and might not function reliably.

Important

Ensure suitable OS driver support exists for your intended PCIe device (host controller) before prototyping.

Connecting a PCIe device follows the standard PCIe convention. CM5 includes on-board AC coupling capacitors for the `PCIe_TX` signals. However, external AC coupling capacitors are required for `PCIe_RX` signals, close to the driving source (the peripheral's TX). PCIe and NVMe cards include these capacitors on board.

To ensure reliable PCIe operation, follow the electrical routing guidance and connect all mandatory control signals outlined below.

2.3.1. Routing guidance

When designing with PCIe on CM5, observe the following conventions for signal routing and connection:

- **Direct IC connection.** If connecting directly to another IC, swap the transmit (TX) and receive (RX) differential pairs; this involves connecting TX to RX and RX to TX. Ensure each receive (`PCIe-Rx`) line has an AC coupling capacitor (220 nF) before it enters the IC.
- **Connector-based connection.** If using a PCIe connector, the signals are labelled from the host's point of view, so TX and RX lines don't need to be swapped.

PCIe differential signals should be routed as 90 Ω differential pairs with proper clearances. Length matching between pairs is unnecessary, but the signals within a pair must be length-matched, ideally within 0.1 mm, to preserve signal integrity. You may swap the positive (P) and negative (N) line within a pair.

2.3.2. Required signals

The following control and clock signals must be handled correctly for proper PCIe operation:

- `PCIe_CLK_nREQ` must be connected to enable clock output from CM5.
- `PCIe_nRST` is required for proper device reset during initialisation or reboot.
- `PCIe_nWAKE` is available, but currently unsupported in software.

2.4. USB interfaces

CM5 provides support for both USB 3.0 (SuperSpeed) and USB 2.0 (High-Speed) interfaces. Both USB 3.0 and USB 2.0 require 90 Ω differential impedance, with length matching within each pair. You may swap the positive (P) and negative (N) signals for USB 3.0 pairs; USB 2.0 pairs can't be P/N swapped.

2.4.1. USB 3.0 (SuperSpeed)

CM5 includes two USB 3.0 interfaces, each supporting up to 5 Gb/s signalling simultaneously. The USB differential pairs should be routed with 90 Ω differential impedance. Length matching between separate pairs is unnecessary, but the P and N signals within each differential pair must be length-matched, ideally within 0.1 mm.

Note

P/N signal swapping is allowed for USB 3.0 pairs.

- `VBUS_EN` Signals to an external power switch to apply power to the USB 3.0 ports.

2.4.2. USB 2.0 (High-Speed)

The USB 2.0 interface supports up to 480 Mb/s signalling. The differential pair should be routed with a 90 Ω differential impedance. The P and N signals within each differential pair should be length-matched, ideally within 0.15 mm.

To enable USB 2.0 functionality, add the `dtoverlay=dwc2,dr_mode=host` overlay setting to your `config.txt` file.

Note

The USB 2.0 port can operate in USB On-The-Go (OTG) mode. While not officially documented, some users have successfully enabled this functionality. The `USB_OTG_ID` pin determines the role (host or device) and is typically connected to the ID pin of a Micro USB connector. To use OTG functionality, it must be enabled in the operating system (OS). For fixed-role use, tie the `USB_OTG_ID` pin to ground.

2.5. Video and display interfaces

CM5 supports a range of high-speed video interfaces for connecting both displays and cameras. It includes two HDMI 2.0 outputs, two 4-lane MIPI interfaces that can be used for DSI displays or CSI cameras, and support for parallel DPI displays through GPIO. CM5 can support up to three simultaneous displays of any type (HDMI, DSI, or DPI).

2.5.1. Dual HDMI 2.0

CM5 includes two HDMI 2.0 interfaces, each capable of supporting a 4K display. Consider the following to ensure reliable HDMI operation:

- HDMI signals must be routed as 100 Ω differential pairs.
 - Within a pair, each signal should be length-matched within 0.15 mm.
 - Between pairs, length matching within 25 mm is sufficient.
- Consumer Electronics Control (CEC) is supported, with an internal 27 k Ω pull-up resistor included in CM5.
- Hotplug Detect (HPD) is supported, with an internal 100 k Ω pull-down resistor included in CM5.
- Extended Display Identification Data (EDID) signals have internal pull-up resistors in CM5.
- Like Raspberry Pi 5, CM5 doesn't have extra ESD protection on HDMI signals because it isn't typically required. Consider whether you might need extra ESD protection and then add it if required.

2.5.2. MIPI (CSI and DSI)

CM5 supports two 4-lane MIPI interfaces for connecting cameras (CSI) and displays (DSI). The MIPI signals must be routed as 100 Ω differential pairs. Within a pair, the signals should be length-matched within 0.15 mm.

In addition to DSI, displays can also be connected using the parallel DPI interface, available using GPIO functions. For more information, see [Section 2.9.2. Alternative GPIO functions](#).

Camera Serial Interface (CSI-2)

CM5 supports camera modules through the CSI interface. For detailed information about the CSI interface, refer to the [Raspberry Pi documentation](#). The following camera sensors are supported by official Raspberry Pi firmware:

- OmniVision OV5647
- Sony IMX219
- Sony IMX296
- Sony IMX477
- Sony IMX708

No security device is required on Compute Module products to use these camera sensors.

Display Serial Interface (DSI)

The DSI interface supports connection to MIPI DSI-compatible displays. CM5 is compatible with displays supported either by:

- The official Raspberry Pi firmware.
- The mainline Linux kernel.

For third-party displays not officially supported, you must provide a custom driver.

2.6. I2C interfaces

CM5 provides two I2C buses that can be repurposed depending on system configuration and peripheral usage.

2.6.1. MIPI I2C bus (SDA0 and SCL0)

The internal I2C bus is normally allocated to the MIPI0 interface. However, it can be used as a general I2C bus or GPIO if the MIPI0 interface isn't in use:

- The serial data pin (SDA0) within the MIPI0 interface is connected to GPIO38 on RP1.
- The serial clock pin (SCL0) within the MIPI0 interface is connected to GPIO39 on RP1.

2.6.2. HAT EEPROM identification I2C bus (ID_SD and ID_SC)

CM5 includes another I2C bus, with signals exposed on the ID_SD (data) and ID_SC (clock) pins. This bus is typically reserved for identifying HATs and controlling MIPI1 devices.

If the firmware isn't using this I2C bus (for example, MIPI1 isn't being used), then these pins can be repurposed as GPIO00 and GPIO01 if needed. When using these pins as GPIO pins, add `force_eeprom_read=0` to the `config.txt` file. This prevents the firmware from checking whether there's a HAT EEPROM available.

2.7. SDIO (CM5Lite only)

This section covers external storage options for **CM5Lite** with the SDIO interface.

CM5Lite doesn't include eMMC storage on board. However, it exposes **Secure Digital Input Output (SDIO)** interface for external storage through the connector: either an external eMMC or SD card (for removable storage).

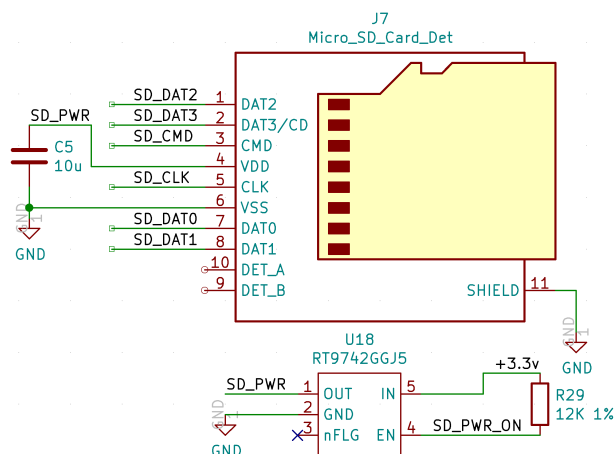
Depending on the type of storage you use, consider the following configuration signals:

- **External eMMC.** Set `SD_VDD_OVERRIDE` to high (CM5_3.3V) to force 1.8 V signalling on the SDIO interface.
- **SD card.** Use the `SD_PWR_ON` signal to control an external power switch for an SD card. To enable SD card boot by default, add a pull-up resistor to keep the power switch on.

Note

SD cards require a power switch controlled by `SD_PWR_ON`; this is the only way to reset the SD card.

Figure 3.
CM5Lite SD card interface



2.8. Debug UART

There is space to fit a debug UART connector for troubleshooting and diagnosing. This connector provides the same functionality as Raspberry Pi 5. The connector is a three-pin, 1 mm pitch JST-SH connector (part number **BM03B-SRSS-TB**). The signals are replicated on the bottom as test points, allowing you access to the debug UART signals even if the main debug connector isn't fitted or available. For information about test points, see [Appendix A. Test Points](#).

2.9. GPIO

There are 28 general-purpose I/O (GPIO) pins available, which correspond to the GPIO pins on the Raspberry Pi 5 40-pin header. These pins have access to internal peripherals, such as DPI, I2C, PWM, SPI, and UART. Details about these features and the available multiplexing options are described in the [RP1 peripherals datasheet](#).

To minimise electromagnetic compatibility (EMC) issues, we recommend setting the drive strength and slew rate to the lowest levels necessary. GPIO2 and GPIO3 include 1.8 kΩ pull-up resistors.

The GPIO bank is powered by the `GPIO_VREF` supply. This can connect to `CM5_1.8V` for 1.8 V signalling, or `CM5_3.3V` for 3.3 V signalling. Don't exceed 50 mA for total current load on all 28 GPIO pins. `GPIO_VREF` must be connected to either `CM5_3.3V` or `CM5_1.8V`. It's possible to use 2.5 V signalling by supplying an external 2.5 V supply to `GPIO_VREF`. This external supply must only be active while `CM5_1.8V` is on and must be fully discharged within 1 ms after `CM5_1.8V` is going low.

2.9.1. Alternative function assignments

Up to nine alternative function assignments are available on the GPIO pins. The following table provides an overview of these alternative functions.

Each GPIO can have only one function at a time. Likewise, each peripheral input (for example, `I2C3_SCL`) must be assigned to only one GPIO pin. If the same peripheral input is connected to multiple GPIOs, the peripheral sees the logical `OR` of these GPIO inputs.

Function selections without a named function in the following table are reserved.

Table 1.

GPIO function selection

GPIO	Function								
	a0	a1	a2	a3	a4	a5	a6	a7	a8
0	SPI0_SIO[3]	DPI_PCLK	UART1_TX	I2C0_SDA		SYS_RIO[0]	PROC_RIO[0]	PIO[0]	SPI2_CSn[0]
1	SPI0_SIO[2]	DPI_DE	UART1_RX	I2C0_SCL		SYS_RIO[1]	PROC_RIO[1]	PIO[1]	SPI2_SIO[1]
2	SPI0_CSn[3]	DPI_VSYNC	UART1_CTS	I2C1_SDA	UART0_IR_RX	SYS_RIO[2]	PROC_RIO[2]	PIO[2]	SPI2_SIO[0]
3	SPI0_CSn[2]	DPI_HSYNC	UART1_RTS	I2C1_SCL	UART0_IR_TX	SYS_RIO[3]	PROC_RIO[3]	PIO[3]	SPI2_SCLK
4	GPCLK[0]	DPI_D[0]	UART2_TX	I2C2_SDA	UART0_RI	SYS_RIO[4]	PROC_RIO[4]	PIO[4]	SPI3_CSn[0]
5	GPCLK[1]	DPI_D[1]	UART2_RX	I2C2_SCL	UART0_DTR	SYS_RIO[5]	PROC_RIO[5]	PIO[5]	SPI3_SIO[1]
6	GPCLK[2]	DPI_D[2]	UART2_CTS	I2C3_SDA	UART0_DCD	SYS_RIO[6]	PROC_RIO[6]	PIO[6]	SPI3_SIO[0]
7	SPI0_CSn[1]	DPI_D[3]	UART2_RTS	I2C3_SCL	UART0_DSR	SYS_RIO[7]	PROC_RIO[7]	PIO[7]	SPI3_SCLK
8	SPI0_CSn[0]	DPI_D[4]	UART3_TX	I2C0_SDA		SYS_RIO[8]	PROC_RIO[8]	PIO[8]	SPI4_CSn[0]
9	SPI0_SIO[1]	DPI_D[5]	UART3_RX	I2C0_SCL		SYS_RIO[9]	PROC_RIO[9]	PIO[9]	SPI4_MISO
10	SPI0_SIO[0]	DPI_D[6]	UART3_CTS	I2C1_SDA		SYS_RIO[10]	PROC_RIO[10]	PIO[10]	SPI4_MOSI
11	SPI0_SCLK	DPI_D[7]	UART3_RTS	I2C1_SCL		SYS_RIO[11]	PROC_RIO[11]	PIO[11]	SPI4_SCLK
12	PWM0[0]	DPI_D[8]	UART4_TX	I2C2_SDA	AUDIO_OUT_L	SYS_RIO[12]	PROC_RIO[12]	PIO[12]	SPI5_CSn[0]
13	PWM0[1]	DPI_D[9]	UART4_RX	I2C2_SCL	AUDIO_OUT_R	SYS_RIO[13]	PROC_RIO[13]	PIO[13]	SPI5_SIO[1]
14	PWM0[2]	DPI_D[10]	UART4_CTS	I2C3_SDA	UART0_TX	SYS_RIO[14]	PROC_RIO[14]	PIO[14]	SPI5_SIO[0]
15	PWM0[3]	DPI_D[11]	UART4_RTS	I2C3_SCL	UART0_RX	SYS_RIO[15]	PROC_RIO[15]	PIO[15]	SPI5_SCLK
16	SPI1_CSn[2]	DPI_D[12]			UART0_CTS	SYS_RIO[16]	PROC_RIO[16]	PIO[16]	
17	SPI1_CSn[1]	DPI_D[13]			UART0_RTS	SYS_RIO[17]	PROC_RIO[17]	PIO[17]	

GPIO	Function								
18	SPI1_CSn[0]	DPI_D[14]	I2S0_SCLK	PWM0[2]	I2S1_SCLK	SYS_RIO[18]	PROC_RIO[18]	PIO[18]	GPCLK[1]
19	SPI1_SIO[1]	DPI_D[15]	I2S0_WS	PWM0[3]	I2S1_WS	SYS_RIO[19]	PROC_RIO[19]	PIO[19]	
20	SPI1_SIO[0]	DPI_D[16]	I2S0_SDI[0]	GPCLK[0]	I2S1_SDI[0]	SYS_RIO[20]	PROC_RIO[20]	PIO[20]	
21	SPI1_SCLK	DPI_D[17]	I2S0_SDO[0]	GPCLK[1]	I2S1_SDO[0]	SYS_RIO[21]	PROC_RIO[21]	PIO[21]	
22	SDIO0_CLK	DPI_D[18]	I2S0_SDI[1]	I2C3_SDA	I2S1_SDI[1]	SYS_RIO[22]	PROC_RIO[22]	PIO[22]	
23	SDIO0_CMD	DPI_D[19]	I2S0_SDO[1]	I2C3_SCL	I2S1_SDO[1]	SYS_RIO[23]	PROC_RIO[23]	PIO[23]	
24	SDIO0_DAT[0]	DPI_D[20]	I2S0_SDI[2]		I2S1_SDI[2]	SYS_RIO[24]	PROC_RIO[24]	PIO[24]	SPI2_CSn[1]
25	SDIO0_DAT[1]	DPI_D[21]	I2S0_SDO[2]	AUDIO_IN_CLK	I2S1_SDO[2]	SYS_RIO[25]	PROC_RIO[25]	PIO[25]	SPI3_CSn[1]
26	SDIO0_DAT[2]	DPI_D[22]	I2S0_SDI[3]	AUDIO_IN_DAT0	I2S1_SDI[3]	SYS_RIO[26]	PROC_RIO[26]	PIO[26]	SPI5_CSn[1]
27	SDIO0_DAT[3]	DPI_D[23]	I2S0_SDO[3]	AUDIO_IN_DAT1	I2S1_SDO[3]	SYS_RIO[27]	PROC_RIO[27]	PIO[27]	SPI1_CSn[1]
29		I2C4_SCL			VBUS_OC	SYS_RIO1[1]	PROC_RIO1[1]		
34	PWM1[2]	GPCLK[3]	VBUS_EN	I2C4_SDA	AUDIO_IN_CLK	SYS_RIO2[0]	PROC_RIO2[0]		
35		PWM1[0]	VBUS_OC	I2C4_SCL	AUDIO_IN_DAT0	SYS_RIO2[1]	PROC_RIO2[1]		
38				I2C6_SDA		SYS_RIO2[4]	PROC_RIO2[4]		
39				I2C6_SCL		SYS_RIO2[5]	PROC_RIO2[5]		
42	GPCLK[5]		VBUS_EN			SYS_RIO2[8]	PROC_RIO2[8]		

2.9.2. Alternative GPIO functions

A variety of alternative GPIO functions accommodate diverse peripheral interfaces and communication protocols. The following list summarises the available peripherals and their supported configurations:

- Five UARTs, with standard and extended wiring options:
 - Four UARTs with 4-wire interfaces for serial communication (TX, RX, CTS, RTS).
 - One UART (UART0) with an 8-wire interface (TX, RX, CTS, RTS, DTR, DCD, DSR, RI) or an IrDA interface (IR_TX, IR_RX).
- One 4-bit SDIO for Secure Digital Input/Output.
- Four PWM channels for pulse-width modulation.
- One I2S Master interface (I2SC0), quadruple lane.
- One I2S, Slave interface (I2SC1), quadruple lane.
- Two AUDIO_OUT PWM audio outputs, which require buffering using a low-noise PSU buffer and filtering with a 22 KHz first-order RC network.
- Two AUDIO_IN digital PDM inputs.
- Two general-purpose clock (GPCLK) outputs.
- One DPI (Display Parallel Interface) with PCLK, DE, VSYNC, HSYNC, and up to 24-bit data.
- 28 GPIO (SYS_RIO) pins.
- Four I2C controllers (SDA, SCL).
- Six SPI controllers, detailed in [Table 2](#), below.

Table 2.
SPI controller configuration

Instance ID	Master/Slave	Chip-select count	Max I/O width
SPI0	M	4	Quad
SPI1	M	3	Dual
SPI2	M	2	Dual
SPI3	M	2	Dual
SPI4	S	1	Single

SPI5	M	2	Dual
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For conventional SPI connections, the Serial Input/Output (SIO) pins numbered 0 and 1 have specific roles:

- `SIO0` is the `MOSI` pin (Master Out Slave In).
- `SIO1` is the `MISO` pin (Master In Slave Out).

The other SIO pins are required for the basic SPI communication to work.

2.9.3. Camera GPIOs (`CAM_GPIO`)

CM5 includes two GPIO control signals for the camera module: `CAM_GPIO0` and `CAM_GPIO1`.

- `CAM_GPIO0` is typically routed to pin 17 on the camera connector. This signal is used to control power to the camera module. `CAM_GPIO0` corresponds to `GPIO34` on RP1.
- `CAM_GPIO1` has been added to CM5 for future expansion, and isn't present on previous Compute Modules; we recommend routing this signal to pin 18 on the camera connector. `CAM_GPIO1` corresponds to `GPIO35` on RP1.

2.10. Status LEDs (`LED_nACT` and `LED_nPWR`)

Status LEDs on CM5 provide visual feedback about the board's activity and power states. These signals replicate the green and red LEDs found on Raspberry Pi 5, helping users monitor eMMC activity, boot errors, and power status.

- `LED_nACT`: This pin drives an LED that indicates eMMC or SD card access, replicating the green LED on Raspberry Pi 5. Under Linux, the LED flashes to signify eMMC or SD card access. If a boot error occurs, the LED flashes an error pattern. To decode these patterns, see the [LED Flash codes](#) in the Raspberry Pi documentation.
- `LED_nPWR`: This pin controls an LED that indicates the board's power status, replicating the red LED on Raspberry Pi 5. When the board is powered but shut down, the LED lights up.

2.11. Fan control (`Fan_PWM` and `Fan_Tacho`)

CM5 provides two pins for monitoring and controlling PWM fans, allowing for fan speed regulation and tachometer feedback.

- `Fan_PWM`: An open-collector output pin designed to drive a variety of PWM-controlled fans.
- `Fan_Tacho`: An input pin with internal pull-up to `CM5_3.3V` for reading the tachometer output signal that is present on many PWM fans; it is connected to `GPIO29` on RP1.

During CM5 shutdown, power to the `Fan_PWM` signal is also stopped. If the fan is powered from a 5 V supply, the fan might still continue to run after power supply shutdown. To prevent this, turn off the supply to the fan simultaneously. For example, you could share power with the external USB ports controlled by `VBUS_EN`. Alternatively, you could use an open-collector buffer (such as a 74LVC1G07) powered from 5 V: connect its input to `CM5_3.3V` and wire the output in parallel with the PWM control line.

2.12. Power management and control

The following signals relate to power state management, power supply negotiation, and system-level control for CM5. Proper use of these signals ensures reliable power sequencing, system startup, shutdown, and battery-backed real-time clock (RTC) operation.

2.12.1. USB-C signals (`CC0` and `CC1`)

The USB-C connector uses the `CC0` and `CC1` Configuration Channel (CC) signals to negotiate power delivery. On CM5, these signals enable the system to request up to 5 V at 5 A from the power source, ensuring efficient and safe power transfer over USB-C.

2.12.2. System control signals (PMIC_EN, PWR_BUT, VBAT, nRPI_BOOT, and EEPROM_nWP)

Table 3 lists key control pins that govern system behaviour during startup, shutdown, and battery-powered operation.

Table 3.

System control signals

Pin	Description	Usage
PMIC_EN	Controls the power-down state of CM5.	Pull low to put CM5 in the lowest power-down state. We recommend only pulling this pin low after OS shutdown.
PWR_BUT	Acts as a power switch when connected to a button, controlling power on and off for CM5.	Pull low briefly to power on or off; hold low for more than 5 seconds to force CM5 to shut down.
VBAT	Supply 2.5 V to 3.5 V to power the on-board RTC. Provides backup power for RTC so that it can keep time even when the board is off.	When CM5 is powered on, RTC uses a small static load. When CM5 is powered off, the load increases to maintain RTC. A typical CR2032 lasts > 3 years when CM5 is unpowered.
nRPI_BOOT	Determines boot source during startup.	Hold low during boot to bypass eMMC and boot through USB 2.0 instead.
EEPROM_nWP	Enables hardware write protection on the EEPROM to prevent data modification.	Pull this pin low to prevent end users from changing the contents of on-board EEPROM. For software configuration instructions, see the EEPROM write protect documentation .

3. Power

CM5 requires a regulated 5 V supply for operation. CM5 can also supply 600 mA at 3.3 V and 1.8 V to peripherals. The following sections describe the required power-up and power-down sequences, typical and maximum power consumption, and the capabilities of the on-board voltage regulators.

3.1. Power-up sequencing

The following list summarises the power-up conditions and sequencing necessary for proper power-up of CM5:

- No pins should be powered before the 5 V rail is active.
- For write-protection of the on-board boot EEPROM, the `EEPROM_nWP` pin must be low before power-up.
- To boot CM5 through USB, the `RPI_nBOOT` pin must be low within 2 ms after the 5 V rail rises.
- The 5 V rail should rise monotonically to at least 4.75 V and remain above this level during operation.
- The power-up sequence begins after the 5 V rail is above 4.75 V and the `PMIC_EN` signal rises.
- The power rails and signals rise in the following order:
 1. 5 V rises
 2. `PMIC_EN` rises
 3. `CM5_+3.3V` rises
 4. `CM5_+1.8V` rises at least 1 ms after `CM5_+3.3V`

3.2. Power-down sequencing

The following list summarises the recommended power-down procedure and considerations for CM5 to ensure safe shutdown and file system integrity:

- To ensure file system consistency, shut down the operating system before removing power.
- If controlled shutdown isn't possible, consider using file systems like `btrfs`, `f2fs`, or `overlayfs`, which can be enabled through `raspi-config`.
- After the operating system has shut down, the 5 V rail can be removed or the `PMIC_EN` pin can be taken low to put the CM5 into the lowest power mode.
- During the shutdown sequence, `CM5_+1.8V` will be discharged before the `CM5_3.3V` rail.

3.3. Power consumption

The exact power consumption of CM5 depends on the tasks being run on it. Typical values are summarised below:

- The lowest shutdown power consumption mode occurs when `PMIC_EN` is driven low, typically around 1.3 mA.
- With `PMIC_EN` high but software shut down, the typical consumption is about 3 mA.
- Idle power consumption is typically 400 mA, but this varies depending on the operating system.
- Operating power consumption is typically around 900 mA, but this depends on the operating system and running tasks.

3.4. Regulator outputs

CM5 has built-in voltage regulators that provide 3.3 V (`CM5_+3.3V`) and 1.8 V (`CM5_+1.8V`) power rails. These regulators can each deliver up to 600 mA of current to external devices or peripherals connected to the board. The current drawn by connected devices from these regulators isn't included in the overall power consumption figures; the reported power usage only accounts for the board itself, not the extra peripherals powered through these regulators.

4. Specifications

This section includes technical descriptions of CM5's components and capabilities, including its dimensions, antenna, pinout, electrical specifications, and thermal characteristics.

4.1. Mechanical specifications

CM5 consists of a compact PCB and an optional wireless antenna. The physical features of CM5's components are summarised in [Section 4.1.1. PCB dimensions](#), which includes information about the module size, PCB thickness, SoC height, and stacking height options. Antenna orientation and clearance requirements are summarised in [Section 4.1.2. Wireless antenna](#).

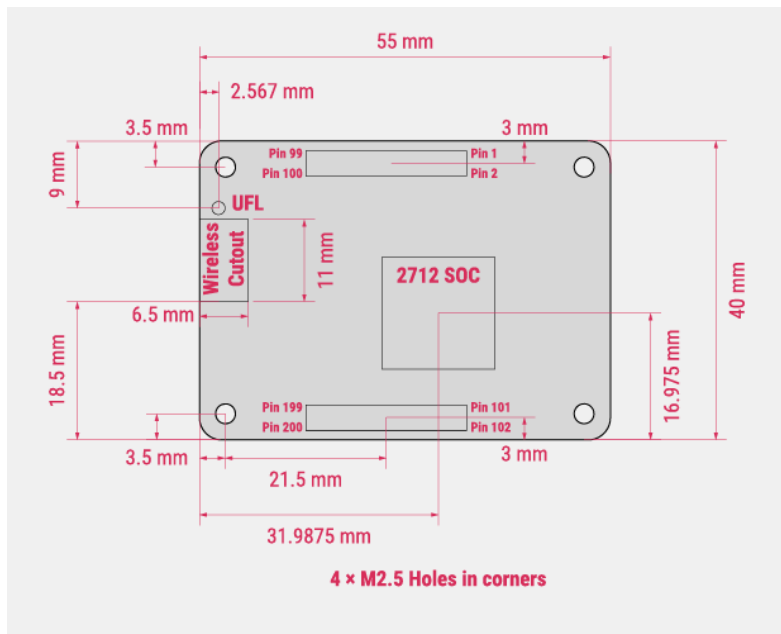
For complete mechanical drawings and CAD models, see the [CM5 design files](#), included in the official design data package. These files are provided to help designers visualise and plan their hardware; they are for reference only and might change over time as the board design is updated or revised.

4.1.1. PCB dimensions

CM5 is a compact 40 mm × 55 mm module. The bare module is 4.6 mm deep; when mounted, the total height becomes either 4.94 mm or 7.44 mm, depending on the chosen stacking height. The mechanical diagram in [Figure 4](#) illustrates the approximate shape and dimensions of CM5 as viewed from the top.

Figure 4.

Mechanical specification of Raspberry Pi Compute Module 5 viewed from the top



Key mechanical specifications of CM5 are as follows:

- **Mounting.** CM5 has four M2.5 mounting holes inset 3.5 mm from the module edge.
- **Thickness.** The PCB is 1.24 mm thick $\pm 10\%$.
- **SoC height.** BCM2712 SoC measures 2.2 mm ± 0.15 mm in height, including solder balls.
- **Stacking height.** Stacking height is determined by the connector used on the carrier board:
 - Amphenol connector part number, 10164227-1001A1RLF, results in a stacking height of 1.5 mm with no clearance underneath the CM5.
 - Amphenol connector part number, 10164227-1004A1RLF, results in a stacking height of 4.0 mm with 2.5 mm clearance underneath the CM5.

The location and arrangement of components on CM5 might change slightly over time due to revisions for cost and manufacturing considerations; however, the maximum component heights and PCB thickness will be kept as specified.

4.1.2. Wireless antenna

If using the on-board PCB antenna, we recommend adhering to the following guidelines:

- Position the on-board wireless antenna so that it faces the edge of the plastic enclosure.
- To avoid degrading wireless performance, ensure any nearby metal includes appropriate cut-outs.
- Maintain a minimum clearance of 10 mm around the PCB antenna; verify actual performance in the final enclosure.
- Don't place any metal, including ground planes, directly beneath the antenna.
- Provide a ground plane cut-out of at least 6.5 mm × 11 mm; ideally, 8 mm × 15 mm or larger.

If these requirements can't be met, wireless performance might be degraded, especially in the 2.4 GHz spectrum. We recommend using the external antenna is used where possible.

For more information about the on-board PCB antenna, see [Section 2.1. Wireless](#).

4.2. Pinout

CM5 includes 200 pins. Each pin is configured with default software function. [Table 4](#) provides a summary of each pin's assignment, including signal names and descriptions.

Table 4.

Pinout for the Raspberry Pi Compute Module 5

Pin	Signal	Description
1	GND	Ground (0 V)
2	GND	Ground (0 V)
3	Ethernet_Pair3_P	Ethernet pair 3 positive (connect to transformer or MagJack)
4	Ethernet_Pair1_P	Ethernet pair 1 positive (connect to transformer or MagJack)
5	Ethernet_Pair3_N	Ethernet pair 3 negative (connect to transformer or MagJack)
6	Ethernet_Pair1_N	Ethernet pair 1 negative (connect to transformer or MagJack)
7	GND	Ground (0 V)
8	GND	Ground (0 V)
9	Ethernet_Pair2_N	Ethernet pair 2 negative (connect to transformer or MagJack)
10	Ethernet_Pair0_N	Ethernet pair 0 negative (connect to transformer or MagJack)
11	Ethernet_Pair2_P	Ethernet pair 2 positive (connect to transformer or MagJack)
12	Ethernet_Pair0_P	Ethernet pair 0 positive (connect to transformer or MagJack)
13	GND	Ground (0 V)
14	GND	Ground (0 V)
15	Ethernet_nLED3	Active-low Ethernet activity indicator (CM5_3.3V signal): typically a green LED is connected to this pin; $I_{OL} = 8 \text{ mA}$ at $V_{OL} < 0.4 \text{ V}$
16	Fan_Tacho	Fan Tacho input pin internally pulled up with a 1.8 kΩ to CM5_3.3V
17	Ethernet_nLED2	Active-low Ethernet speed indicator (CM5_3.3V signal): typically a yellow LED is connected to this pin; a low state indicates the 1 Gbit or 100 Mbit link; $I_{OL} = 8 \text{ mA}$ at $V_{OL} < 0.4 \text{ V}$
18	Ethernet_SYNC_OUT	IEEE1588 SYNC output pin, may be configured to be an input (CM5_3.3V signal; $I_{OL} = 8 \text{ mA}$ at $V_{OL} < 0.4 \text{ V}$)
19	Fan_PWM	Open drain output
20	EEPROM_nWP	Leave floating NB internally pulled up to CM5_3.3V through 100 kΩ ($V_{IL} < 0.8 \text{ V}$); can be grounded to prevent writing to the on-board EEPROM that stores the boot code

Pin	Signal	Description
21	LED_nACT	Active-low Pi activity LED; 20 mA max, 5 V tolerant ($V_{OL} < 0.4$ V); this signal drives the green LED on Raspberry Pi 5
22	GND	Ground (0 V)
23	GND	Ground (0 V)
24	GPIO26	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
25	GPIO21	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
26	GPIO19	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
27	GPIO20	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
28	GPIO13	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
29	GPIO16	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
30	GPIO6	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
31	GPIO12	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
32	GND	Ground (0 V)
33	GND	Ground (0 V)
34	GPIO5	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
35	ID_SC	(RP1 GPIO 1) GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
36	ID_SD	(RP1 GPIO 0) GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
37	GPIO7	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
38	GPIO11	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
39	GPIO8	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
40	GPIO9	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
41	GPIO25	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
42	GND	Ground (0 V)
43	GND	Ground (0 V)
44	GPIO10	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
45	GPIO24	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
46	GPIO22	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
47	GPIO23	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
48	GPIO27	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
49	GPIO18	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
50	GPIO17	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
51	GPIO15	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
52	GND	Ground (0 V)
53	GND	Ground (0 V)
54	GPIO4	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
55	GPIO14	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V
56	GPIO3	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V; internal 1.8 k Ω pull-up to GPIO_VREF
57	SD_CLK	SD card clock signal (only available on CM5Lite)
58	GPIO2	GPIO: typically a 3.3 V signal, but can be a 1.8 V signal by connecting GPIO_VREF to CM5_1.8V; internal 1.8 k Ω pull-up to GPIO_VREF

Pin	Signal	Description
59	GND	Ground (0 V)
60	GND	Ground (0 V)
61	SD_DAT3	SD card/eMMC Data3 signal (only available on CM5Lite)
62	SD_CMD	SD card/eMMC Command signal (only available on CM5Lite)
63	SD_DAT0	SD card/eMMC Data0 signal (only available on CM5Lite)
64	SD_DAT5	SD card/eMMC Data5 signal (only available on CM5Lite)
65	GND	Ground (0 V)
66	GND	Ground (0 V)
67	SD_DAT1	SD card/eMMC Data1 signal (only available on CM5Lite)
68	SD_DAT4	SD card/eMMC Data4 signal (only available on CM5Lite)
69	SD_DAT2	SD card/eMMC Data2 signal (only available on CM5Lite)
70	SD_DAT7	SD card/eMMC Data7 signal (only available on CM5Lite)
71	GND	Ground (0 V)
72	SD_DAT6	SD card/eMMC Data6 signal (only available on CM5Lite)
73	SD_VDD_OVERRIDE	Connect to CM5_3.3V to force SD card/eMMC interface to 1.8 V signalling instead of 3.3 V, otherwise leave unconnected. Typically only used if external eMMC is connected.
74	GND	Ground (0 V)
75	SD_PWR_ON	Output to power switch for the SD card; CM5 sets this pin high (3.3 V) to signal that power to the SD card should be turned on; internally pulled up to CM5_3.3V with a 4.53 kΩ resistor (only available on CM5Lite)
76	VBAT	RTC battery input 2.5 V to 3.5 V; typically 3 V
77	5V (Input)	4.75 V to 5.25 V main power input
78	GPIO_VREF	Must be connected to CM5_3.3V (pins 84 and 86) for 3.3 V GPIO0-27 or CM5_1.8V (pins 88 and 90) for 1.8 V GPIO0-27; this pin can't be floating or connected to ground
79	5V (Input)	4.75 V to 5.25 V; main power input
80	SCL0	I2C clock pin (GPIO39): typically used for camera and display; internal 1.8 kΩ pull-up to CM5_3.3V
81	5V (Input)	4.75 V to 5.25 V; main power input
82	SDA0	I2C data pin (GPIO38): typically used for camera and display; internal 1.8 kΩ pull-up to CM5_3.3V
83	5V (Input)	4.75 V to 5.25 V; main power input
84	CM5_3.3V (Output)	3.3 V ± 5%. Power output max 300 mA per pin for a total of 600 mA; powered down during power-off or when PMIC_Enable set low
85	5V (Input)	4.75 V to 5.25 V; main power input
86	CM5_3.3V (Output)	3.3 V ± 5%. Power output max 300 mA per pin for a total of 600 mA; powered down during power-off or when PMIC_Enable set low
87	5V (Input)	4.75 V to 5.25 V; main power input
88	CM5_1.8V (Output)	1.8 V ± 5%. Power output max 300 mA per pin for a total of 600 mA; powered down during power-off or when PMIC_Enable set low
89	WL_nDisable	Can be left floating; if driven low, the Wi-Fi interface will be disabled. Internally pulled up through 1.8 kΩ to CM5_3.3V
90	CM5_1.8V (Output)	1.8 V ± 5%. Power output max 300 mA per pin for a total of 600 mA; powered down during power-off or when PMIC_Enable set low
91	BT_nDisable	Can be left floating; if driven low, the Bluetooth interface will be disabled; internally pulled up through 1.8 kΩ to CM5_3.3V
92	PWR_Button	Pull low to force power off or power on from previous software powered off state; internally pulled up to 5 V through 10 kΩ

Pin	Signal	Description
93	nRPIBOOT	A low on this pin forces booting from an RPI server (for example, PC or a Raspberry Pi); if not used, leave floating; internally pulled up through 10 kΩ to CM5_3.3V
94	CC1	USB PSU PD signal; wire to a USB-C connector to enable 5 A at 5 V negotiation.
95	LED_nPWR	3.3 V signal: active-low output to drive Power On LED; this signal needs to be buffered
96	CC2	USB PSU PD signal; wire to a USB-C connector to enable 5 A at 5 V negotiation.
97	CAM_GPI00	3.3 V signal: can be a GPIO (GPIO34) or part of the bus with pin 100
98	GND	Ground (0 V)
99	PMIC_Enable	Input; drive low to power off CM5 internally pulled up with a 100 kΩ to 5 V
100	CAM_GPI01	3.3 V signal (GPIO35): internally pulled up with 15 kΩ to CM5_3.3V
101	USB_OTG_ID	Input (3.3 V signal): USB OTG pin; internally pulled up; when grounded, CM5 becomes a USB host but the correct OS driver must also be used
102	PCIE_CLK_nREQ	Input (3.3 V signal): PCIe clock request pin (low to request PCI clock); internally pulled up
103	USB_N	USB 2.0 D-
104	PCIE_nWAKE	3.3 V signal: PCIe WAKE# signal can be left unconnected if wake up isn't required; internally pulled up
105	USB_P	USB 2.0 D+
106	PCIE_PWR_EN	3.3 V signal: active high, used to signal that a PCIe device can be powered down when low
107	GND	Ground (0 V)
108	GND	Ground (0 V)
109	PCIE_nRST	Output (3.3 V signal): PCIe reset active-low
110	PCIE_CLK_P	PCIe clock-out positive (100 MHz)
111	VBUS_EN	3.3 V signal: active high to signal USB 3.0 ports should be powered. This is driven internally by GPIO42 from RP1
112	PCIE_CLK_N	PCIe clock out negative (100 MHz)
113	GND	Ground (0 V)
114	GND	Ground (0 V)
115	MIPI0_D0_N	MIPI0 D0 negative
116	PCIE_RX_P	Input PCIe GEN 2 RX positive NB external AC coupling capacitor required
117	MIPI0_D0_P	MIPI0 D0 positive
118	PCIE_RX_N	Input PCIe GEN 2 RX negative NB external AC coupling capacitor required
119	GND	Ground (0 V)
120	GND	Ground (0 V)
121	MIPI0_D1_N	MIPI0 D1 negative
122	PCIE_TX_P	Output PCIe GEN 2 TX positive NB AC coupling capacitor included on CM5
123	MIPI0_D1_P	MIPI0 D1 positive
124	PCIE_TX_N	Output PCIe GEN 2 TX positive NB AC coupling capacitor included on CM5
125	GND	Ground (0 V)
126	GND	Ground (0 V)
127	MIPI0_C_N	MIPI0 clock negative
128	USB3-0-RX_N	USB 3.0 port 0 RX input negative
129	MIPI0_C_P	MIPI0 clock positive
130	USB3-0-RX_P	USB 3.0 port 0 RX input positive
131	GND	Ground (0 V)

Pin	Signal	Description
132	GND	Ground (0 V)
133	MIPI0_D2_N	MIPI0 D2 negative
134	USB3-0-DP	USB 3.0 port 0 DP
135	MIPI0_D2_P	MIPI0 D2 positive
136	USB3-0-DM	USB 3.0 port 0 DM
137	GND	Ground (0 V)
138	GND	Ground (0 V)
139	MIPI0_D3_N	MIPI0 D3 negative
140	USB3-0-TX_N	USB 3.0 port 0 TX output negative NB AC coupling capacitor included on CM5
141	MIPI0_D3_P	MIPI0 D3 positive
142	USB3-0-TX_P	USB 3.0 port 0 TX output positive NB AC coupling capacitor included on CM5
143	HDMI1_HOTPLUG	Input HDMI1 hotplug; internally pulled down with a 100 kΩ. 5 V tolerant. Can be connected directly to a HDMI connector.
144	GND	Ground (0 V)
145	HDMI1_SDA	Bidirectional HDMI1 SDA; internally pulled up with a 1.8 kΩ. 5 V tolerant; can be connected directly to a HDMI connector
146	HDMI1_TX2_P	Output HDMI1 TX2 positive
147	HDMI1_SCL	Bidirectional HDMI1 SCL; internally pulled up with a 1.8 kΩ. 5 V tolerant; can be connected directly to a HDMI connector
148	HDMI1_TX2_N	Output HDMI1 TX2 negative
149	HDMI1_CEC	Input HDMI1 CEC; internally pulled up with a 27 kΩ. 5 V tolerant; can be connected directly to a HDMI connector
150	GND	Ground (0 V)
151	HDMI0_CEC	Input HDMI0 CEC; internally pulled up with a 27 kΩ. 5 V tolerant; can be connected directly to a HDMI connector
152	HDMI1_TX1_P	Output HDMI1 TX1 positive
153	HDMI0_HOTPLUG	Input HDMI0 hotplug; internally pulled down 100 kΩ. 5 V tolerant; can be connected directly to a HDMI connector
154	HDMI1_TX1_N	Output HDMI1 TX1 negative
155	GND	Ground (0 V)
156	GND	Ground (0 V)
157	USB3-1-RX_N	USB 3.0 port 1 RX input negative
158	HDMI1_TX0_P	Output HDMI1 TX0 positive
159	USB3-1-RX_P	USB 3.0 port 1 RX input positive
160	HDMI1_TX0_N	Output HDMI1 TX0 negative
161	GND	Ground (0 V)
162	GND	Ground (0 V)
163	USB3-1-DP	USB 3.0 port 1 DP
164	HDMI1_CLK_P	Output HDMI1 clock positive
165	USB3-1-DM	USB 3.0 port 1 DM
166	HDMI1_CLK_N	Output HDMI1 clock negative
167	GND	Ground (0 V)
168	GND	Ground (0 V)

Pin	Signal	Description
169	USB3-1-TX_N	USB 3.0 Port 1 TX output negative NB AC coupling capacitor included on CM5
170	HDMI0_TX2_P	Output HDMI0 TX2 positive
171	USB3-1-TX_P	USB 3.0 Port 1 TX output positive NB AC coupling capacitor included on CM5
172	HDMI0_TX2_N	Output HDMI0 TX2 negative
173	GND	Ground (0 V)
174	GND	Ground (0 V)
175	MIPI1_D0_N	MIPI1 D0 negative
176	HDMI0_TX1_P	Output HDMI0 TX1 positive
177	MIPI1_D0_P	MIPI1 D0 positive
178	HDMI0_TX1_N	Output HDMI0 TX1 negative
179	GND	Ground (0 V)
180	GND	Ground (0 V)
181	MIPI1_D1_N	MIPI1 D1 negative
182	HDMI0_TX0_P	Output HDMI0 TX0 positive
183	MIPI1_D1_P	MIPI1 D1 positive
184	HDMI0_TX0_N	Output HDMI0 TX0 negative
185	GND	Ground (0 V)
186	GND	Ground (0 V)
187	MIPI1_C_N	MIPI1 clock negative
188	HDMI0_CLK_P	Output HDMI0 clock positive
189	MIPI1_C_P	MIPI1 clock positive
190	HDMI0_CLK_N	Output HDMI0 clock negative
191	GND	Ground (0 V)
192	GND	Ground (0 V)
193	MIPI1_D2_N	MIPI1 D2 negative
194	MIPI1_D3_N	MIPI1 D3 negative
195	MIPI1_D2_P	MIPI1 D2 positive
196	MIPI1_D3_P	MIPI1 D3 positive
197	GND	Ground (0 V)
198	GND	Ground (0 V)
199	HDMI0_SDA	Bidirectional HDMI0 SDA; internally pulled up with a 1.8 kΩ. 5 V tolerant; can be connected directly to an HDMI connector
200	HDMI0_SCL	Bidirectional HDMI0 SCL; internally pulled up with a 1.8 kΩ. 5 V tolerant; can be connected directly to an HDMI connector

4.2.1. Pin guidelines

The following instructions provide guidance for grounding, connector usage, voltage limits, and power rail considerations, and precautions against improper voltage application.

- **Grounding.** Always connect all ground pins on any connector in use. If none of the signals on the second connector (pins 101 to 200) are in use, then you may omit the connector (including its ground pins) to reduce costs; however, omitting the second connector can affect mechanical stability.
- **GPIO voltage limits.** GPIO pins 0 to 27 are the same as the 40-pin connector on Raspberry Pi 5. Depending on your signalling, their voltage must not exceed:

- 3.3 V (CM5_3.3V) when using 3.3 V signalling.
- 1.8 V (CM5_1.8V) when using 1.8 V signalling.
- **Power rails.** If you use power rails CM5_3.3V or CM5_1.8V to supply devices other than the GPIO reference voltage (GPIO_VREF), you must design for safe behaviour during unexpected power loss (for example, the 5 V line falls below 4.5 V):
 - If you use the 1.8 V rail (CM5_1.8V), ensure that the current draw goes down to zero (no load) if power suddenly drops.
 - If you use the 3.3 V rail (CM5_3.3V), ensure that the 3.3 V rail voltage never falls below the 1.8 V rail voltage if power suddenly drops. The 3.3 V rail voltage usually stays above the 1.8 V rail voltage during power-down, but verify your design. If the 3.3 V rail does fall below 1.8 V, add circuitry to disconnect 3.3 V devices to prevent damage.
- **Reverse voltage.** Don't apply reverse voltage on any pin. This means that when CM5 is powered-down or off, there must be no external voltage applied to any pin, otherwise CM5 might not power up again.

4.2.2. Differential pairs

We recommend that positive and negative (P/N) signals within a differential pair are length-matched to within 0.15 mm. Depending on the interface, the matching tolerance can be more relaxed between different pairs. For example, HDMI pair-to-pair matching can typically be within 25 mm, so no extra matching is required on a typical board.

100 Ω differential pair signal lengths

All 100 Ω differential pairs on CM5 are length-matched to less than 0.05 mm for P/N signals. We recommend that pairs are also matched on the interface board. On CM5, pair-to-pair length matching isn't always maintained because many interfaces don't require precise matching between different pairs. Table 5 lists the track-length differences within each differential pair group on CM5. A non-zero value represents how much longer in millimetres (mm) that track is when compared to the signal with zero length difference in the group.

Table 5.

100 Ω differential pair signal lengths

Signal	Length
MIPI0	
MIPI0_C_N	0.78
MIPI0_C_P	0.78
MIPI0_D0_N	0.01
MIPI0_D0_P	0.02
MIPI0_D1_N	0.4
MIPI0_D1_P	0.4
MIPI0_D2_N	0.04
MIPI0_D2_P	0.03
MIPI0_D3_N	0.01
MIPI0_D3_P	0
MIPI1	
MIPI1_C_N	1.28
MIPI1_C_P	1.27
MIPI1_D0_N	0
MIPI1_D0_P	0
MIPI1_D1_N	1.06
MIPI1_D1_P	1.05
MIPI1_D2_N	0.83
MIPI1_D2_P	0.84
MIPI1_D3_N	3.79
MIPI1_D3_P	3.79

Signal	Length
HDMI0	
HDMI0_CLK_N	0.91
HDMI0_CLK_P	0.91
HDMI0_TX0_N	0.18
HDMI0_TX0_P	0.18
HDMI0_TX1_N	0
HDMI0_TX1_P	0
HDMI0_TX2_N	0.25
HDMI0_TX2_P	0.25
HDMI1	
HDMI1_CLK_N	2.99
HDMI1_CLK_P	2.99
HDMI1_TX0_N	4.76
HDMI1_TX0_P	4.75
HDMI1_TX1_N	5.18
HDMI1_TX1_P	5.18
HDMI1_TX2_N	0
HDMI1_TX2_P	0
Ethernet	
Ethernet_Pair0_P	2.92
Ethernet_Pair0_N	2.93
Ethernet_Pair1_P	0
Ethernet_Pair1_N	0
Ethernet_Pair2_P	0.59
Ethernet_Pair2_N	0.60
Ethernet_Pair3_P	0.38
Ethernet_Pair3_N	0.38

90 Ω differential pair signal lengths

All 90 Ω differential pairs on CM5 (including USB pairs) are length-matched to less than 0.01 mm for P/N signals. USB 3.0 pairs don't require pair-to-pair matching within a port group. We recommend that pairs are also matched on the interface board. On CM5, pair-to-pair length matching isn't always maintained because many interfaces don't require precise matching between different pairs. [Table 6](#) lists the track-length differences within each differential pair group on CM5. A non-zero value represents how much longer in millimetres (mm) that track is when compared to the signal with zero length difference in the group.

Table 6.

90 Ω differential pair signal lengths

Signal	Length
PCIe_CLK_P	0.00
PCIe_CLK_N	0.01
PCIe_TX_P	3.71
PCIe_TX_N	3.72
PCIe_RX_P	0.84

PCIe_RX_N	0.84
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4.3. Electrical specifications

For safe and reliable operation of CM5, observe the following electrical parameters and limitations.

4.3.1. Absolute maximum ratings

Warning

Stresses above those listed in [Table 7](#) can cause permanent damage to the device. This is a stress rating only; functional operation of the device under these or any other conditions above those listed in the operational sections of this specification isn't implied. Exposure to absolute maximum rating conditions for extended periods can affect device reliability.

[Table 7](#) lists the absolute maximum ratings for key voltage parameters on the CM5. These values represent the limits beyond which damage to the device can occur and shouldn't be exceeded.

Table 7.

Absolute maximum ratings

Symbol	Parameter	Minimum	Maximum	Unit
V_{IN}	5 V input voltage	-0.5	6.0	V
V_{GPIO_VREF}	GPIO voltage	-0.5	3.6	V
V_{gpio}	GPIO input voltage	-0.5	$V_{GPIO_VREF} + 0.5$	V

V_{GPIO_VREF} is the GPIO bank voltage, which must be tied to either the 3.3 V or the 1.8 V rail of CM5.

4.3.2. DC characteristics

[Table 8](#) details the DC electrical characteristics of the GPIO pins on CM5. It describes how the GPIO pins perform under different conditions (such as different reference voltages) and provides the expected ranges for each parameter (minimum, typical, and maximum values). For the electrical details of other interfaces in CM5, see [Section 2. Interfaces](#).

Table 8.

DC characteristics

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
$V_{IL(gpio)}$	Input low voltage	$V_{GPIO_VREF} = 3.3\text{ V}$	0	-	0.8	V
$V_{IH(gpio)}$	Input high voltage	$V_{GPIO_VREF} = 3.3\text{ V}$	2.0	-	V_{GPIO_VREF}	V
$V_{IL(gpio)}$	Input low voltage	$V_{GPIO_VREF} = 2.5\text{ V}$	0	-	0.7	V
$V_{IH(gpio)}$	Input high voltage	$V_{GPIO_VREF} = 2.5\text{ V}$	1.7	-	V_{GPIO_VREF}	V
$V_{IL(gpio)}$	Input low voltage	$V_{GPIO_VREF} = 1.8\text{ V}$	0	-	$0.35 \cdot V_{GPIO_VREF}$	V
$V_{IH(gpio)}$	Input high voltage	$V_{GPIO_VREF} = 1.8\text{ V}$	$0.65 \cdot V_{GPIO_VREF}$	-	V_{GPIO_VREF}	V
$I_{IL(gpio)}$	Input leakage current	$V_{GPIO_VREF} = 3.3\text{ V}$	-	-	3	μA
$I_{IL(gpio)}$	Input leakage current	$V_{GPIO_VREF} = 2.5\text{ V}$	-	-	5	μA
$I_{IL(gpio)}$	Input leakage current	$V_{GPIO_VREF} = 1.8\text{ V}$	-	-	7	μA
$V_{OL(gpio)}$	Output low voltage	-	-	-	0.4	V
$V_{OH(gpio)}$	Output high voltage	$V_{GPIO_VREF} = 3.3\text{ V}$	$V_{GPIO_VREF} - 0.4$	-	-	V
$V_{OH(gpio)}$	Output high voltage	$V_{GPIO_VREF} = 2.5\text{ V}$	$V_{GPIO_VREF} - 0.5$	-	-	V

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
$V_{OH(gpio)}$	Output high voltage	$V_{GPIO_VREF} = 1.8\text{ V}$	$V_{GPIO_VREF} - 0.4$	-	-	V
$I_{OL(gpio)}$	Output current	2 mA, $V_{GPIO_VREF} = 3.3\text{ V}$	6.1	9.6	13.5	mA
$I_{OL(gpio)}$	Output current	4 mA, $V_{GPIO_VREF} = 3.3\text{ V}$	9.2	14.3	20.2	mA
$I_{OL(gpio)}$	Output current	8 mA, $V_{GPIO_VREF} = 3.3\text{ V}$	15.3	23.9	33.7	mA
$I_{OL(gpio)}$	Output current	12 mA, $V_{GPIO_VREF} = 3.3\text{ V}$	18.4	28.7	40.5	mA
$I_{OH(gpio)}$	Output current	2 mA, $V_{GPIO_VREF} = 3.3\text{ V}$	4.5	6.3	8.4	mA
$I_{OH(gpio)}$	Output current	4 mA, $V_{GPIO_VREF} = 3.3\text{ V}$	6.8	9.5	12.6	mA
$I_{OH(gpio)}$	Output current	8 mA, $V_{GPIO_VREF} = 3.3\text{ V}$	11.4	15.8	21	mA
$I_{OH(gpio)}$	Output current	12 mA, $V_{GPIO_VREF} = 3.3\text{ V}$	13.6	19	25.2	mA
$I_{OL(gpio)}$	Output current	2 mA, $V_{GPIO_VREF} = 2.5\text{ V}$	4.7	8	12.2	mA
$I_{OL(gpio)}$	Output current	4 mA, $V_{GPIO_VREF} = 2.5\text{ V}$	7.1	12	18.2	mA
$I_{OL(gpio)}$	Output current	8 mA, $V_{GPIO_VREF} = 2.5\text{ V}$	11.8	20	30.4	mA
$I_{OL(gpio)}$	Output current	12 mA, $V_{GPIO_VREF} = 2.5\text{ V}$	14.1	24	36.4	mA
$I_{OH(gpio)}$	Output current	2 mA, $V_{GPIO_VREF} = 2.5\text{ V}$	3.5	5.1	7	mA
$I_{OH(gpio)}$	Output current	4 mA, $V_{GPIO_VREF} = 2.5\text{ V}$	5.2	7.6	10.5	mA
$I_{OH(gpio)}$	Output current	8 mA, $V_{GPIO_VREF} = 2.5\text{ V}$	8.7	12.7	17.6	mA
$I_{OH(gpio)}$	Output current	12 mA, $V_{GPIO_VREF} = 2.5\text{ V}$	10.4	15.2	21.1	mA
$I_{OL(gpio)}$	Output current	2 mA, $V_{GPIO_VREF} = 1.8\text{ V}$	4.4	8.1	13.6	mA
$I_{OL(gpio)}$	Output current	4 mA, $V_{GPIO_VREF} = 1.8\text{ V}$	8.8	16.3	27.2	mA
$I_{OL(gpio)}$	Output current	8 mA, $V_{GPIO_VREF} = 1.8\text{ V}$	11.8	21.7	36.3	mA
$I_{OL(gpio)}$	Output current	12 mA, $V_{GPIO_VREF} = 1.8\text{ V}$	16.2	29.2	49.9	mA
$I_{OH(gpio)}$	Output current	2 mA, $V_{GPIO_VREF} = 1.8\text{ V}$	3.4	5.3	7.7	mA
$I_{OH(gpio)}$	Output current	4 mA, $V_{GPIO_VREF} = 1.8\text{ V}$	6.9	10.5	15.4	mA
$I_{OH(gpio)}$	Output current	8 mA, $V_{GPIO_VREF} = 1.8\text{ V}$	9.1	14	20.6	mA
$I_{OH(gpio)}$	Output current	12 mA, $V_{GPIO_VREF} = 1.8\text{ V}$	12.6	19.3	28.3	mA
$R_{PU(gpio)}$	Pull-up resistor	$V_{GPIO_VREF} = 3.3\text{ V}$	37	55	86	k Ω
$R_{PD(gpio)}$	Pull-down resistor	$V_{GPIO_VREF} = 3.3\text{ V}$	35	55	98	k Ω
$R_{PU(gpio)}$	Pull-up resistor	$V_{GPIO_VREF} = 2.5\text{ V}$	49	77	123	k Ω
$R_{PD(gpio)}$	Pull-down resistor	$V_{GPIO_VREF} = 2.5\text{ V}$	49	84	155	k Ω
$R_{PU(gpio)}$	Pull-up resistor	$V_{GPIO_VREF} = 1.8\text{ V}$	38	64	106	k Ω
$R_{PD(gpio)}$	Pull-down resistor	$V_{GPIO_VREF} = 1.8\text{ V}$	58	103	189	k Ω

4.3.3. Current consumption

Table 9 presents key current consumption characteristics for CM5 under various operating conditions. It details the typical shutdown, idle, operational, and RTC currents measured with different input voltages and control signals. Actual figures greatly depend on the end application.

Table 9.

Current consumption characteristics for Raspberry Pi Compute Module 5 (CM5)

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
$I_{shutdown}$	Shutdown current	$PMIC_ENABLE < 0.4\text{ V}$	-	1.3	-	mA
$I_{shutdown}$	Shutdown current	$PMIC_ENABLE > 2\text{ V}$	-	3	-	mA

I_{idle}	Idle current	PMIC_ENABLE > 2 V	-	400	-	mA
I_{load}	Operation current	PMIC_ENABLE > 2 V	-	900	-	mA
I_{VBAT}	RTC current	Vin = 5 V	-	1.7	-	μA
I_{VBAT}	RTC current	Vin = 0 V	-	6	-	μA

4.4. Thermal characteristics

CM5 contains less metal in the PCB and fewer connectors than Raspberry Pi 5, which means that it has less passive heat sinking than Raspberry Pi 5.

The BCM2712 SoC on CM5 has built-in thermal management that reduces its clock speed to keep the SoC temperature below 85°C. To avoid overheating, the SoC might automatically throttle its performance in high ambient temperatures. If the SoC can't reduce its temperature enough through throttling, its case temperature can exceed 85°C. Any thermal management solution must ensure that the ambient temperatures of the other silicon components on the board stay within their safe operating range.

CM5's overall operating temperature range is from -20°C to +85°C (non-condensing). Wireless RF performance is best within -20°C to +75°C.

4.5. Mean time between failure (MTBF)

Mean time between failure (MTBF) measures how long, on average, each device is expected to operate before failure. [Table 10](#) shows the MTBF for CM5 and CM5Lite, which varies depending on environmental conditions.

Table 10.

Mean time between failure (MTBF) for Raspberry Pi Compute Module 5 (CM5)

Environment	Description	CM5 MTBF	CM5Lite MTBF
Ground, benign	A stable, non-mobile environment where temperature and humidity are controlled, such as laboratories, business and scientific computer complexes, and medical equipment rooms. In these environments, devices generally last longer.	143 000 hours	168 000 hours
Ground, mobile	A high-stress environment with vibration, temperature swings, humidity variations, and frequent movement, such as equipment in vehicles and handheld communication devices. In these environments, life expectancy drops.	16 000 hours	16 000 hours

5. Troubleshooting

CM5 has a number of power-up and boot stages before it starts. If an error occurs at any of these stages, CM5 might fail to start or run as expected. The following sections help you to diagnose and resolve the issue by:

- Checking hardware power rails and signals for proper voltages and load behaviour.
- Verifying bootloader firmware operation and enabling diagnostics or alternate boot modes.
- Managing EEPROM firmware updates and write protection for boot reliability.

We also recommend avoiding known issues by ensuring the system software (firmware and kernel) are up to date. Keeping your firmware up to date can resolve many system issues and improve stability because newer versions contain improvements to the system. Similarly, new kernel releases often include important security patches and performance improvements.

5.1. Hardware power rails

CM5 requires stable power to start up. Check key power rails (5 V, 3.3 V, and 1.8 V) to verify power enable signals and identify any back-feeding caused by external devices or wiring.

1. **Test the 5 V supply under load:** Pull `PMIC_EN` low and apply an external 2 A load to the 5 V supply. The voltage should remain above 4.75 V (including noise), ideally, staying above 4.9 V.
2. **Check for back-feeding on 3.3 V and 1.8 V rails:** Remove the external 2 A load, but keep `PMIC_EN` pulled low. If the voltage for 3.3 V and 1.8 V exceeds 200 mV, there might be an external power path back-feeding the board, possibly through digital pins such as Ethernet.
3. **Check `PMIC_EN` goes high:** Remove the pull-down on `PMIC_EN` and then check that `PMIC_EN` now goes high; measure the voltage on this pin or check its logic state to confirm it goes high.
4. **Confirm that the voltage rails rise correctly:**
 - Check the 3.3 V supply rises to more than 3.15 V. If it doesn't, this suggests there is too much load on the 3.3 V rail.
 - Check the 1.8 V supply rises to more than 1.71 V. If it doesn't, this suggests there is too much load on the 1.8 V rail.
5. **Check the activity LED (`LED_nACT`) to verify the boot process:** The LED should oscillate to indicate booting; check it isn't flashing an error code. To decode error code patterns, see the [LED Flash codes](#) in the Raspberry Pi documentation.

5.2. Bootloader firmware

The bootloader firmware manages the initial startup of CM5. If your CM5 fails to start correctly, verify bootloader operation, then enable diagnostic modes. To verify bootloader operation:

- Connect an HDMI cable. The bootloader has started and is running correctly if the HDMI diagnostics screen appears.
- Connect a USB serial cable to GPIO pins 14 and 15. This allows you to receive bootloader output logs (through UART), which help verify what stage the bootloader is at and diagnose any issues. For more information, see [Configure UARTs](#).

If the bootloader isn't running as expected, enable diagnostic modes: short the `nRPIBOOT` pin to ground to force USB boot mode. The CM5IO board has a jumper for `nRPIBOOT` that you can use to enable different boot modes (for example, network boot) and UART logging. For more information, see [Flash an image to a Compute Module](#).

5.3. EEPROM management and firmware updates

For reliable startup and system stability on your CM5, keep the bootloader EEPROM up to date, including correct management of write protection.

- **Check EEPROM write-protection.** The on-board EEPROM can be write-protected by shorting the `EEPROM_nWP` pin to ground. The CM5IO board provides a jumper for `EEPROM_nWP` to enable or disable write protection.
- **If necessary, update or repair EEPROM.** CM5 won't run `recovery.bin` from the eMMC (or SD card on CM5Lite); update or repair the bootloader EEPROM on your CM5 through `usbboot` or self-update. Ensure write protection is disabled before attempting to update the EEPROM. For more information, see [Boot EEPROM](#).

6. Ordering information

CM5 comes in a range of variants distinguished by wireless capability, RAM size, and eMMC storage capacity. Each CM5 variant is identified by a unique order code (part number). The available product variants are detailed in the tables below. Custom configurations can also be arranged to suit specific requirements.

6.1. Order quantity and packaging

You can order a specific number of one or more CM5 devices that will arrive individually boxed, or you can make a bulk order that will come in a single shipper. Small quantities supplied in individual cardboard boxes have an internal ESD coating so that a separate ESD bag isn't required. This packaging is recyclable to reduce waste.

6.2. Part number codes

[Table 11](#) explains the structure of part numbers for CM5 variants. It details how the model, wireless capability, RAM size, and eMMC storage capacity are encoded within the part number.

Table 11.

Part number information for Raspberry Pi Compute Module 5 (CM5)

Model	Wireless	RAM LPDDR4x	eMMC storage
CM5	0 = No	01 = 1 GB	000 = 0 GB (Lite)
	1 = Yes	02 = 2 GB	008 = 8 GB
		04 = 4 GB	016 = 16 GB
		08 = 8 GB	032 = 32 GB
		16 = 16 GB	064 = 64 GB
			128 = 128 GB
Example part number			
CM5	1	02	032

6.3. Product variants

[Table 12](#) shows available variants for CM5 by part number, detailing wireless support, RAM size, eMMC storage capacity and RPL numbers. Other configurations can be custom ordered.

Table 12.

Available product variants for Raspberry Pi Compute Module 5 (CM5)

Part number	Wireless	RAM LPDDR4x	Storage eMMC	RPL number
CM5002000	-	2 GB	Lite (0 GB)	SC1556
CM5002016	-	2 GB	16 GB	SC1558
CM5002032	-	2 GB	32 GB	SC1559
CM5002064	-	2 GB	64 GB	SC1560
CM5102000	Yes	2 GB	Lite (0 GB)	SC1586
CM5102016	Yes	2 GB	16 GB	SC1588
CM5102032	Yes	2 GB	32 GB	SC1589
CM5102064	Yes	2 GB	64 GB	SC1590

Part number	Wireless	RAM LPDDR4x	Storage eMMC	RPL number
CM5004000	-	4 GB	Lite (0 GB)	SC1562
CM5004016	-	4 GB	16 GB	SC1564
CM5004032	-	4 GB	32 GB	SC1565
CM5004064	-	4 GB	64 GB	SC1566
CM5104000	Yes	4 GB	Lite (0 GB)	SC1592
CM5104016	Yes	4 GB	16 GB	SC1594
CM5104032	Yes	4 GB	32 GB	SC1595
CM5104064	Yes	4 GB	64 GB	SC1596
CM5008000	-	8 GB	Lite (0 GB)	SC1568
CM5008016	-	8 GB	16 GB	SC1570
CM5008032	-	8 GB	32 GB	SC1571
CM5008064	-	8 GB	64 GB	SC1572
CM5108000	Yes	8 GB	Lite (0 GB)	SC1598
CM5108016	Yes	8 GB	16 GB	SC1600
CM5108032	Yes	8 GB	32 GB	SC1601
CM5108064	Yes	8 GB	64 GB	SC1602
CM5016000	-	16 GB	Lite (0 GB)	SC1574
CM5016016	-	16 GB	16 GB	SC1576
CM5016032	-	16 GB	32 GB	SC1577
CM5016064	-	16 GB	64 GB	SC1578
CM5116000	Yes	16 GB	Lite (0 GB)	SC1604
CM5116016	Yes	16 GB	16 GB	SC1606
CM5116032	Yes	16 GB	32 GB	SC1607
CM5116064	Yes	16 GB	64 GB	SC1608

Appendix A. Test Points

CM5 contains test points: pins on the board that you can use to power the board, program it, or debug it without needing to use the main 100-pin connectors.

A.1. Test point map

Table 13 lists the coordinates (X and Y) of test points on CM5 and what each test point is used for. Most signals replicate pins on the 100-pin connectors of CM5.

Table 13.

Test points for Raspberry Pi Compute Module 5 (CM5)

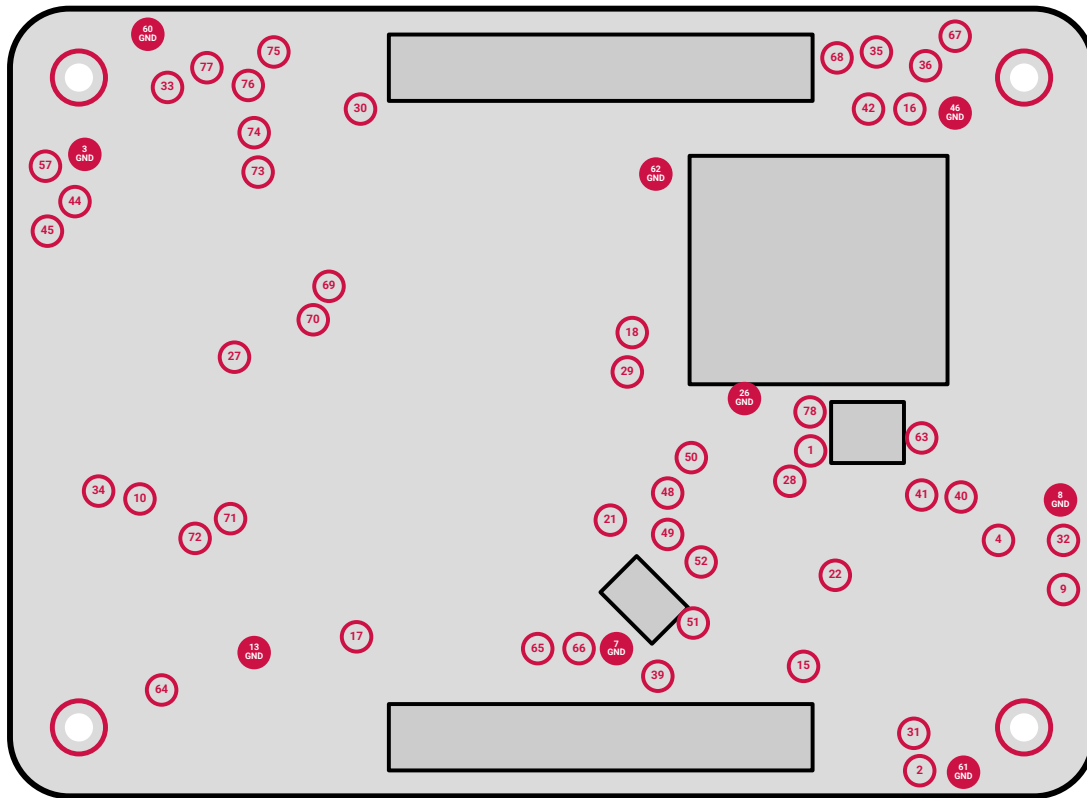
Reference	X	Y	Name
MH4	51.5	36.5	Mounting Hole
MH3	3.5	36.5	Mounting Hole
MH2	51.5	3.5	Mounting Hole
MH1	3.5	3.5	Mounting Hole
TP1	14.34	17.54	5 V
TP2	8.8	1.3	RUN
TP3	51.2	32.6	GND
TP4	4.8	13	reserved
TP7	24.2	7.5	GND
TP8	1.65	15.05	GND
TP9	1.5	10.5	reserved
TP10	48.4	15.1	reserved
TP13	42.6	7.3	GND
TP15	14.7	6.6	reserved
TP16	9.3	34.9	nRPIB00T
TP17	37.4	8.1	reserved
TP18	23.4	23.55	reserved
TP21	24.5125	14.025	nRESET_OUT
TP22	13.0875	11.225	reserved
TP26	17.7	20.2	GND
TP27	43.6	22.3	reserved
TP28	15.4	16	reserved
TP29	23.65	21.55	reserved
TP30	37.2	34.9	reserved
TP31	9.1	3.2	reserved
TP32	1.5	13	reserved
TP33	47	36	CM5_3V3
TP34	50.5	15.5	CM5_1V8
TP35	11	37.8	DEBUG_UART_TX
TP36	8.5	37.1	DEBUG_UART_RX
TP39	22.1	6.1	reserved

Reference	X	Y	Name
TP40	6.7	15.2	reserved
TP41	8.7	15.3	reserved
TP42	11.4	34.9	PWR_BUT
TP44	51.7	30.2	reserved
TP45	53.1	28.7	reserved
TP46	7	34.7	GND
TP48	21.6	15.4	SOC_TRST_N
TP49	21.6	13.3	SOC_TDI
TP50	20.4	17.2	SOC_TDO
TP51	20.3	8.8	SOC_TMS
TP52	19.9	11.9	SOC_TCK
TP57	53.2	32	reserved
TP60	48	38.7	GND
TP61	6.575	1.225	GND
TP62	22.2	31.6	GND
TP63	8.7	18.2	5v_Sense
TP64	47.3	5.4	reserved
TP65	28.2	7.5	USBC_D_N
TP66	26.1	7.5	USBC_D_P
TP67	7	38.6	LED_nPWR
TP68	13	37.5	LED_nACT
TP69	38.8	25.9	ETH0_P
TP70	39.6	24.2	ETH0_N
TP71	43.8	14.1	ETH1_N
TP72	45.6	13.1	ETH1_P
TP73	42.4	31.7	ETH2_P
TP74	42.6	33.7	ETH2_N
TP75	41.6	37.8	ETH3_P
TP76	42.9	36.1	ETH3_N
TP77	45	37	GPIO_VREF
TP78	14.37	19.52	5 V

Figure 5 shows the location of these test points on the back of the board.

Figure 5.

The back of Raspberry Pi Compute Module 5 (CM5) showing the location of the test points.



A.2. Test point connections for power and programming

Use the following test points to power, program, and boot CM5 without using the main 100-pin connectors:

- **Power (Vin).** Use test points TP1 and TP78 to supply 5 V power to CM5. At a minimum, connect ground to test points TP26, TP61, and TP8. If possible, use more ground points.
- **Debug UART.** Test points TP35 (TX) and TP36 (RX) provide serial debug communication lines. Use TP46 for ground. This is useful for programming and debugging during boot.
- **Raspberry Pi boot mode.** Pins TP65 and TP66 serve as USB data lines. Connect TP7 as ground, and also ground TP16 (nRPI_BOOT) to force the board into Raspberry Pi boot mode.
- **Ethernet boot.** Connect test points TP69 and TP76 to an external Ethernet MagJack (the Ethernet connector with magnetics).

Appendix B. CM4 and CM5 differences

This section describes the differences between Raspberry Pi Compute Module 5 (CM5) and the previous module, Raspberry Pi Compute Module 4 (CM4).

B.1. Pinout changes

CM5 introduces specific pin-level changes from CM4, including updated pin functions and signal repurposing to support new hardware features.

B.1.1. Per-pin differences

[Table 14](#) compares the exact per-pin changes between CM4 and CM5. It highlights updated functions and signal repurposing to reflect the new hardware capabilities and interfaces.

Table 14.

Pin changes between Raspberry Pi Compute Module 4 (CM4) and Raspberry Pi Compute Module 5 (CM5)

Pin	CM4	CM5	Details
16	SYNC_IN	Fan_tacho	Fan tachometer input
19	Ethernet nLED1	Fan_PWM	Fan PWM output
76	Reserved	VBAT	RTC battery (there's a constant load of a few uA even if CM5 is powered)
92	RUN_PG	PWR_Button	Replicates the power button on Raspberry Pi 5: a short press signals that the device should wake up or shut down; a long press forces shutdown
93	nRPIBOOT	nRPIBOOT	
94	AnalogIP1	CC1	Connects to the CC1 line of a USB-C connector so the PMIC to negotiate 5 A
96	AnalogIP0	CC2	Connects to the CC2 line of a USB-C connector so the PMIC to negotiate 5 A
99	Global_EN	PMIC_ENABLE	No external change
100	nEXTRST	CAM_GPIO1	Pulled up on CM5, but driven low during boot to emulate a nRESET signal
104	Reserved	PCIE_DET_nWAKE	PCIE nWAKE; pull-up to CM5_3v3 with an 8.2 kΩ
106	Reserved	PCIE_PWR_EN	Signals if the PCIe device can be powered up or down; active high
111	VDAC_COMP	VBUS_EN	Output to signal USB VBUS should be enabled
128	CAM0_D0_N	USB3-0-RX_N	May be P/N swapped
130	CAM0_D0_P	USB3-0-RX_P	May be P/N swapped
134	CAM0_D1_N	USB3-0-DP	USB 2.0 signal
136	CAM0_D1_P	USB3-0-DM	USB 2.0 signal
140	CAM0_C_N	USB3-0-TX_N	May be P/N swapped
142	CAM0_C_P	USB3-0-TX_P	May be P/N swapped
157	DSI0_D0_N	USB3-1-RX_N	May be P/N swapped
159	DSI0_D0_P	USB3-1-RX_P	May be P/N swapped
163	DSI0_D1_N	USB3-1-DP	USB 2.0 signal
165	DSI0_D1_P	USB3-1-DM	USB 2.0 signal
169	DSI0_C_N	USB3-1-TX_N	May be P/N swapped
171	DSI0_C_P	USB3-1-TX_P	May be P/N swapped

B.1.2. Summary of functional and hardware changes

In addition to the above, there have been broader design changes. The following list summarises the functional consequences of these changes, as well as the impact of some of the above pin changes:

- **Connectors.** The connectors have changed brand and have been tested to higher currents to support CM5.
- **Thickness.** The PCB for CM5 is 0.04 mm thicker than CM4, but the main processor is thinner.
- **PCIe clock.** PCIe CLK signals are no longer capacitively coupled.
- **ESD protection.** CM4 has extra ESD protection on the HDMI, SDA, SCL, HPD, and CEC signals. This is removed from CM5.
- **Dual-purpose DSI and CSI signals.** `CAM1` and `DSI1` signals became dual-purpose and can be used for either a CSI camera or a DSI display. For more information, see [Section 2.5. Video and display interfaces](#).
- **USB ports.** For more information about USB ports on CM5, see [Section 2.4. USB interfaces](#).
 - The `CAM0` port (pins 128 to 142) on CM4 is a USB 3.0 port on CM5.
 - The `DSI0` port (pins 157 to 171) on CM4 is a USB 3.0 port on CM5.
- **VBUS enable pin.** Pin 111 (`VDAC_COMP` on CM4) has been repurposed as a VBUS enable pin controlling power to the two USB 3.0 ports on CM5. Power to the USB 3.0 ports is enabled when the pin is active high.
- **PD CC signals.** Pins 94 and 96, the two ADC channels on CM4, have become the Power Delivery (PD) Configuration Channel (CC) signals within the USB-C connector.

B.2. Track lengths

CM5 has updated HDMI and Ethernet track lengths compared to CM4. These changes improve pair-to-pair skew and remain well within tolerances, so no functional impact is expected for previous Compute Modules.

- **HDMI0:** P/N pairs remain length-matched, but the skew between pairs is now less than 1 mm. This is unlikely to make a difference because the skew between pairs can be up to 25 mm on previous Compute Modules.
- **HDMI1:** P/N pairs remain length-matched, but the skew between pairs is now less than 5 mm. This is unlikely to make a difference because the skew between pairs can be up to 25 mm on previous Compute Modules.
- **Ethernet:** P/N pairs remain length-matched, but the skew between pairs is now less than 4 mm. This is unlikely to make a difference because the skew between pairs can be up to 12 mm on previous Compute Modules.

B.3. Power budget

CM5 delivers significantly more performance than CM4, and therefore consumes more power. Power supply designs should accommodate 5 V at up to 2.5 A. If this creates an issue with an existing board design, lowering the CPU clock rate can reduce the peak power consumption.

For more information about power requirements for CM5, see [Section 3.3. Power consumption](#).

Appendix C. Documentation history

Date	Changes
8 June, 2026	Add a testpoint diagram. Minor additions and fixes.
6 October, 2025	Minor additions and fixes.
28 August, 2025	Updated structure, grammar, and wording for clarity and style. Made minor corrections. Added information about 16 GB memory and 64 GB storage eMMC. Added information about CM5 connectors. Updated diagrams.
21 July, 2025	Update to new company format.
27 November, 2024	Initial release of Raspberry Pi Compute Module 5 (CM5).



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